



Model Curriculum

JUNIOR SOFTWARE

SECTOR: IT-ITeS

SUB-SECTOR: IT Services

OCCUPATION: Application Development

REFERENCE ID: SSC/Qo5o8, version 1.0

NSQF LEVEL: 4



Certificate

COMPLIANCE TO QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the
IT-ITeS Sector Skills Council NASSCOM

for

MODEL CURRICULUM

Complying to the National occupation standards of
Job Role / Qualification Pack - Junior Software Developer
QP No. SSC/Q0508

Date of Issuance: December 1st 2016

Valid Upto *: December 1st 2017

* Valid up to the next review date of the Qualification Pack



Authorised Signatory

(IT-ITeS Sector Skills Council NASSCOM)

Table of Contents

Curriculum/Syllebus	4
Module 1: Basics of IT	5
Module 2: Problem Solving and Program Design	5
Module 3: Self and work Management	5
Module 4: Team Work and Communication	6
Module 5: Managing Health and Safety	6
Module 6: Data and Information Management	6
Module 7: Learning and Self Development	7
Unique Equipment Required	7
Assessment Criteria	10
Trainer Prerequisites for Job role: "Qualification Pack: "SSC/Q0508"	14

Junior Software Developer

CURRICULUM / SYLLABUS

This program is aimed at training candidates for the job of “Junior Software Developer” in the “IT-ITes” Sector/Industry and aims at building the following key competencies in the learner.

Program Name	Junior Software Developer		
Qualification Pack Name & Reference ID.	Junior Software Developer SSC/Q0508, version 1.0		
Version No.	1.0	Version Update Date	31/12/2015
Pre-requisites to Training	10 th Standard		
Training Outcomes	After completing this programme, participants will be able to: • assist in performing software construction and software testing entry-level tasks in the IT Services industry • manage work to meet requirements • maintain a healthy, safe and secure working environment		

The Course encompasses all six National Occupational Standards (NOS) of “Junior Software Developer” SSC/Qo508 Qualification Pack issued by “IT-ITes Sector Skills Council NASSCOM”.

Sr. No.	Module	Key Learning Outcomes	Equipment Required
1	Basics of IT Theory Duration (hh:mm) 5:00 Practical Duration (hh:mm) 15:00 Corresponding NOS Code SSC/No506	Candidates will be able to: Demonstrate basic computer and internet literacy including operating a computer, describing its major components and how they work, using Windows and Linux OS, operating a browser, searching the internet, managing mails and using social internet media.	Refer to Unique Equipment Required section
2	Problem Solving and Program Design Theory Duration (hh:mm) 30:00 Practical Duration (hh:mm) 60:00 Corresponding NOS Code SSC/No506	Candidates will be able to: - Demonstrate aptitude for analysing information and making logical conclusions. - Demonstrate knowledge of the foundational mathematical concepts in computing.	Refer to Unique Equipment Required section
3	Basic Algorithms and Application Development Theory Duration (hh:mm) 30:00 Practical Duration (hh:mm) 60:00 Corresponding NOS Code SSC/No506	Candidates will be able to: - Design algorithms to solve problems and convert them into code using the appropriate programming language constructs. - Read and execute a test case and record the outcome in the appropriate template. - Communicate effectively with appropriate people w.r.t. assigned roles in simple English – both oral and written.	Refer to Unique Equipment Required section
4	Self and work Management Theory Duration (hh:mm) 30:00 Practical Duration (hh:mm) 70:00 Corresponding NOS Code	Candidates will be able to: - Establish and agree work requirements with appropriate people - Keep immediate work area clean and tidy - Utilize time effectively - Use resources correctly and efficiently - Treat confidential information correctly - Work in line with organization’s policies and procedures - Work within the limits of job role	Refer to Unique Equipment Required section

Sr. No.	Module	Key Learning Outcomes	Equipment Required
	SSC/N9001	- Obtain guidance from appropriate people, where necessary - Ensure work meets the agreed requirements	
5	Team Work and Communication Theory Duration (hh:mm) 12:00 Practical Duration (hh:mm) 38:00 Corresponding NOS Code SSC/N9002	Candidates will be able to: - Obtain guidance from appropriate people to agree the analysis to be performed on the data - Obtain advice and guidance from appropriate people on issues with data analysis outside their area of competence or - Review the results of their analysis with appropriate people - Undertake modifications to your analysis based on inputs from appropriate people - Communicate with colleagues clearly, concisely and accurately - Work with colleagues to integrate their work effectively with them - Pass on essential information to colleagues in line with organizational requirements - Work in ways that show respect for colleagues - Carry out commitments they have made to colleagues - Let colleagues know in good time if they cannot carry out your commitments, explaining the reasons - Identify any problems they have working with colleagues and take the initiative to solve these problems - Follow the organization's policies and procedures for working with colleagues	Refer to Unique Equipment Required Section
6	Managing Health and Safety Theory Duration (hh:mm) 12:00 Practical Duration (hh:mm) 38:00 Corresponding NOS Code SSC/N9003	Candidates will be able to: - Comply with organization's current health, safety and security policies and procedures - Report any identified breaches in health, safety, and security policies and procedures to the designated person - Identify and correct any hazards that can deal with safely, competently and within the limits of authority - Report any hazards that one is not competent to deal with to the relevant person in line with organizational procedures and warn other people who may be affected - Follow their organization's emergency procedures promptly, calmly, and efficiently - Identify and recommend opportunities for improving health, safety, and security to the designated person - Complete any health and safety records legibly and accurately	Refer to Unique Equipment Required section
7	Data and Information Management Theory Duration	Candidates will be able to:	Refer to Unique Equipment Required Section

Sr. No.	Module	Key Learning Outcomes	Equipment Required
	(hh:mm) 15:00 Practical Duration (hh:mm) 35:00 Corresponding NOS Code SSC/9004	- Establish and agree with appropriate people the data/information they need to provide, the formats in which you need to provide it, and when they need to provide it - Obtain the data/information from reliable sources - Check that the data/information is accurate, complete and up-to-date - Obtain advice or guidance from appropriate people where there are problems with the data/information - Carry out rule-based analysis of the data/information, if required - Insert the data/information into the agreed formats - Check the accuracy of work, involving colleagues where required - Report any unresolved anomalies in the data/information to appropriate people - Provide complete, accurate and up-to-date data/information to the appropriate people in the required formats on time	
8	Learning and Self Development Theory Duration (hh:mm) 5:00 Practical Duration (hh:mm) 20:00 Corresponding NOS Code SSC/N9005	Candidates will be able to: - Obtain advice and guidance from appropriate people to develop your knowledge, skills and competence - Identify accurately the knowledge and skills they need for your job role - Identify accurately their current level of knowledge, skills and competence and any learning and development needs - Agree with appropriate people a plan of learning and development activities to address their learning needs - Undertake learning and development activities in line with their plan - Apply new knowledge and skills in the workplace, under supervision - Obtain feedback from appropriate people on their knowledge and skills and how effectively you apply them - Review their knowledge, skills and competence regularly and take appropriate action	Refer to Unique Equipment Required Section
	Total Duration: Theory Duration (hh:mm) 114:00 Practical Duration (hh:mm) 286:00	Unique Equipment Required: Training room should be fully furnished with the following equipment / tools / accessories. Additional / specific resources, wherever applicable (e.g. Hardware, software) are indicated in the main text corresponding to relevant learning outcome. For Domain NOS, For NOS SSC/No506 – HTML, C++ / Java, IDE General:	

Sr. No.	Module	Key Learning Outcomes	Equipment Required
		- Comfortable seats with adequate lighting, controlled temperature and acoustics for training and learning - White Board, Markers and Eraser - Projector with screen - Flip chart with markers - Faculty's PC/Laptop with latest configuration and internet connection - Supporting software / applications for projecting audio, video, recording, - Presentation Tools to support learning activities: - Intranet - Email - IMs - Learning management system e.g. Moodle, Blackboard to enable blended learning - Microphone / voice system for lecture and class activities - Handy Camera - Stationery kit – Staples, Glue, Chart Paper, Sketch Pens, Paint Box, Scale, A4 Sheets - For IT Lab sessions: Computer Lab with 1:1 PC : trainee ratio and having internet connection, MS Office / Open office, Browser, Outlook / Any other Email Client and chat tools. - Assessment and Test Tools for day to day online Tests and Assessments - For team discussions: Adequate seating arrangement in full / half circle format for one or more teams as per planned team composition. - Reading Resources: Access to relevant sample documents and learning forums to enable self-study before and after each training session.	

Grand Total Course Duration: 400 Hours o Minutes

Notes from IT-ITes Sector Skills Council NASSCOM

1. This document outlines the broad scope of coverage. This should be linked with OBF and training delivery plan. OBF (Outcome based framework) reflects the pedagogy used to ensure an expected outcome. Training delivery plan focuses on the sequence of delivery.
2. Though many NOSs have some seemingly common outcomes, notably core/generic, professional and technical skills, it is imperative to understand the contextual difference between them. For example, writing skills required to document program structure and code (in SSC/No506) are different from the writing skills required to prepare a time plan (in SSC/No001). Training providers are advised to,
 - a. Embed such skills development in the learning pedagogy for each expected outcome
 - b. Prepare a detailed session plan for training delivery with focus on sequence and duration of training
 - c. Run a diagnostic test to assess prior learning of students and help trainers / students identify the need for gap training, optimal duration and suitable training methodology. Accordingly, more introductory level sessions may be included in guided or self-paced mode of learning. E.g. adding some sessions on Functional English or Use of Internet and MS Office.

Trainer Prerequisites for Job role: "Junior Software Developer" mapped to Qualification Pack: "SSC/Qo5o8"

Sr. No.	Area	Details
1	Job Description	To deliver accredited training service, mapping to the curriculum detailed above, in accordance with the Qualification Pack SSC/Qo5o8.
2	Personal Attributes	Aptitude for conducting training, and pre/ post work to ensure competent, employable candidates at the end of the training. Strong communication skills, interpersonal skills, ability to work as part of a team; a passion for quality and for developing others; well-organised and focused, eager to learn and keep oneself updated with the latest in this field.
3	Minimum Educational Qualifications	Minimum 12 th Standard; Preferred Master's degree in any discipline
4a	Domain Certification	Minimum accepted score in SSC Assessment is 90% per NOS being taught in QP SSC/Qo5o8. Additional certification in customer orientation, dealing with difficult customers, written communication etc. will be an added advantage.
4b	Platform Certification	Recommended that the Trainer is certified for the Job Role: "Trainer" mapped to the Qualification Pack: "SSC/Q14o2". Minimum accepted score is 70% per NOS.
5	Experience	Field experience: Minimum 2 years' experience in the same domain Training experience: 1 year preferred

Annexure: Assessment Criteria

Assessment Criteria for Junior Software Developer	
Job Role	Junior Software Developer
Qualification Pack	SSC/Qo5o8
Sector Skill Council	IT-ITes

Sr. No.	Guidelines for Assessment
1	Criteria for assessment for each Qualification Pack (QP) will be created by the Sector Skill Council (SSC). Each performance criteria (PC) will be assigned Theory and Skill/Practical marks proportional to its importance in NOS.
2	The assessment will be conducted online through assessment providers authorised by SSC.
3	Format of questions will include a variety of styles suitable to the PC being tested such as multiple choice questions, fill in the blanks, situational judgment test, simulation and programming test.
4	To pass a QP, a trainee should pass each individual NOS. Standard passing criteria for each NOS is 70%.
5	For latest details on the assessment criteria, please visit www.sscnasscom.com .

Assessment Outcome (Nos Code And Description)		Assessment criteria (PC)	Total Marks	Out Of	Marks Allocation Theory Skills Practical	
1.SSC/No5o6 (Deal remotely with customer queries - Domestic)	PC1. greet customers and verify details, following your organization’s procedures		120	12.5	2.5	10
	PC2. read carefully, summarize, and obtain customer confirmation of, your understanding of queries			12.5	2.5	10
	PC3. express your concern for any difficulties caused and your commitment to resolving queries			15	0	15
	PC4. record and categorize queries accurately using your organization’s query management tool			5	0	5
	PC5. refer queries outside your area of competence or authority promptly to appropriate people			2.5	0	2.5
	PC6. access your organization’s knowledge base for solutions to queries, where available			2.5	0	2.5
	PC7. resolve queries within your area of competence or authority in line with organizational guidelines and service level agreements (SLAs)			15	0	15
	PC8. obtain advice and guidance from appropriate people, where necessary			2.5	0	2.5
	PC9. obtain confirmation from customers that queries have been resolved to satisfaction			10	0	10
	PC10. record the resolution of queries accurately using your organization’s query management tool			35	15	20
	PC11. comply with relevant standards, policies, procedures and guidelines when dealing remotely with customer queries			7.5	0	7.5

		NOS Total	120	20	100
2.SSC/N9001 (Manage your work to meet requirements)	PC1. establish and agree your work requirements with appropriate people	40			
	PC2. keep your immediate work area clean and tidy		10	5	5
	PC3. utilize your time effectively		5	0	5
	PC4. use resources correctly and efficiently		5	5	0
	PC5. treat confidential information correctly		5	2.5	2.5
	PC6. work in line with your organization's policies and procedures		5	0	5
	PC7. work within the limits of your job role		2.5	0	2.5
	PC8. obtain guidance from appropriate people, where necessary		2.5	0	2.5
	PC9. ensure your work meets the agreed requirements		2.5	0	2.5
		NOS Total	40	12.5	27.5
3.SSC/N9003 (Maintain a healthy, safe and secure working environment)	PC1. comply with your organization's current health, safety and security policies and procedures	40	10	5	5
	PC2. report any identified breaches in health, safety, and security policies and procedures to the designated person		5	0	5
	PC3. identify and correct any hazards that you can deal with safely, competently and within the limits of your authority		10	5	5
	PC4. report any hazards that you are not competent to deal with to the relevant person in line with organizational procedures and warn other people who may be affected		5	0	5
	PC5. follow your organization's emergency procedures promptly, calmly, and efficiently		5	0	5
	PC6. identify and recommend opportunities for improving health, safety, and security to the designated person		2.5	0	2.5
	PC7. complete any health and safety records legibly and accurately		2.5	0	2.5
		NOS Total	40	10	30